

Updates & Notes

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Why Inverse Dynamics $\rightarrow$ Forward Kinematics Pipeline

Learning ID in latent space is (probably) easier than learning FK

- ▶ Invertibility $\rightarrow$ bijective
- ▶ bijective $\rightarrow$ *no* information loss
 - ▶ inputs can be reconstructed from outputs
- ▶ Inspired by [[@brandfonbrener2023inverse](#)]

Bounds on FK Errors

- ▶ Function of ID errors and Lipschitz of FK?

Total Pipeline:

- ▶ Learn ID (good)
- ▶ Get embeddings from ID (good)
- ▶ Train FK on embeddings (good)
- ▶ Train decoder on embeddings (?)
 - ▶ Currently, decoder has kind of high loss, but only with an early-stopped ID embedding

Upcoming

- ▶ Better exploration policy
 - ▶ Random? Brownian noise in action space?
- ▶ Planning

WCP

► ~ halfway there